

Evaluation Workshop for Speech and Language Technologies (EWSLT)

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Why EWSLT?

Despite significant investment and growing interest in Indian language technologies, progress remains fragmented and hard to measure. There is no single platform today where ASR, TTS, MT, Indic LLMs, SLMs, and spoken translation systems can be evaluated side by side using standardized benchmarks. Every team builds its own test sets, picks its own metrics, and reports results in its own way — making it nearly impossible to compare systems fairly or track real progress over time. As a result, advancements in the Indian speech and language space appear slower and less coherent than they actually are. Good work gets buried because there's no common ground to demonstrate it on.

At the same time, global benchmarks that the research community relies on — LibriSpeech, FLORES, CommonVoice — were never designed for the Indian context. They don't account for code-switching between Hindi and English in the same sentence, the morphological richness of Dravidian languages, the diversity of scripts from Devanagari to Ol Chiki, or the noisy mobile-first audio that real Indian users produce. Evaluating Indian systems on these benchmarks is like measuring a fish by its ability to climb a tree — the results tell you very little about actual performance in deployment.

EWSLT exists to fix this. One umbrella bringing together all major speech and language tasks, with benchmarks built ground-up for Indian languages, new evaluation measures that go beyond WER and BLEU, and coverage that extends to low-resource and underserved languages — not just the top five or ten. The goal is simple: give India a standardized, high-quality evaluation foundation so that researchers, startups, and institutions can build, compare, and improve on common ground.

Goals of EWSLT

- Develop and release comprehensive benchmarks covering the full spectrum of Indian speech and language technologies — ASR, TTS, MT, Indic LLMs, SLMs, and SSMT — all under one unified banner
- Introduce new evaluation measures for TTS, ASR, and SSMT that go beyond conventional metrics like WER, BLEU, and MOS, which were never designed for Indian language complexities

- Design benchmarks ground-up for Indian contexts — accounting for code-switching, dialectal variation, script diversity, morphological richness, and noisy real-world audio
- Extend benchmark coverage to low-resource and underserved Indian languages, not just the top five or ten
- Establish a standardized, high-quality evaluation foundation that the entire community — researchers, startups, labs, and industry — can adopt as the common standard
- Enable fair, reproducible comparison across systems so that real progress in Indian language technologies becomes measurable and visible

Why Participate in EWSLT?

- **One place for all Indian languages** — a single, unified platform to evaluate your systems across Indian ASR, TTS, MT, LLMs, and SLMs instead of scattered, inconsistent test sets
- **Leaderboards** — see where your system stands against others on standardized Indian language tasks, with transparent and comparable rankings
- **Benchmarks that reflect reality** — evaluation on code-switching, dialectal variation, noisy audio, and real-world conditions that global benchmarks ignore
- **New evaluation measures** — go beyond standard WER/BLEU/MOS with metrics designed specifically for Indian language nuances
- **Visibility for your work** — published results on a recognized Indian benchmark carry weight with the research community, funding bodies, and industry
- **Shape the standard** — the evaluation criteria defined now will influence research and product directions for years; participating gives you a seat at that table
- **Low-resource language coverage** — contribute to and benefit from evaluation infrastructure for languages that have almost no existing benchmarks
- **Reduce duplicated effort** — stop building internal test sets from scratch; use shared, curated, high-quality benchmarks and spend your time on the actual models
- **Community and collaboration** — connect with researchers, startups, and labs across India working on the same problems, share insights, and identify partnership opportunities

Challenges Overview

Speech Challenges

ASR

Challenge 1: ASR Evaluation Measures

- Goal: Propose and validate new evaluation metrics for Indian ASR that go beyond WER — capturing semantic impact of errors, code-switching accuracy, morphological variation, and domain-specific correctness
- Output: New metric definitions, implementations, and validation studies demonstrating stronger correlation with real-world ASR usability for Indian languages

Challenge 2: Code-Switched ASR

- Goal: Build ASR systems that accurately recognize speech where speakers mix languages within and across sentences — Hindi-English, Tamil-English, Telugu-Hindi, and other Indian code-switching pairs
- Output: ASR models evaluated on overall transcription accuracy, switch-point accuracy, and word-level language identification on code-switched Indian speech

Challenge 3: Domain-Specific ASR

- Goal: Evaluate and improve ASR performance on specialized Indian domains like agriculture, healthcare, legal, banking, and education where domain vocabulary and speaking styles differ from general-purpose data
- Output: Domain-adapted ASR systems with benchmarked accuracy on domain-specific Indian language test sets

Challenge 4: Low-Resource ASR — Doing More with Less

- Goal: Maximize ASR performance for Indian languages given strictly limited labeled data, using techniques like self-supervised pretraining, data augmentation, transfer learning, semi-supervised methods, or synthetic data generation
- Output: Reproducible techniques and trained models demonstrating competitive accuracy under tight data constraints

Challenge 5: Long-Form ASR

- Goal: Evaluate ASR systems on extended Indian language audio — lectures, meetings, podcasts, court proceedings — where systems must maintain accuracy over long durations with speaker turns, topic shifts, and contextual dependencies
 - Output: ASR systems benchmarked on long-form Indian language audio with metrics capturing accuracy degradation over time
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TTS

Challenge 6: Evaluation Metrics for TTS

- Goal: Design and validate new automatic and human evaluation metrics that separately and meaningfully assess naturalness, prosodic accuracy, and intelligibility for Indian language TTS — moving beyond subjective MOS scores
- Output: New metric definitions, toolkits, and validation results showing reliability and reproducibility across Indian languages

Challenge 7: Multilingual TTS

- Goal: Build TTS systems that serve multiple Indian languages from a single model while matching or approaching monolingual quality across languages
- Output: Multilingual TTS models evaluated on cross-lingual voice consistency, script handling, and quality across high-resource and low-resource Indian languages

Challenge 8: TTS for Indian Languages

- Goal: Evaluate TTS systems specifically on the unique phonological and prosodic challenges of Indian languages — schwa deletion, nasalization, gemination, and intonation patterns that vary across language families and dialects
- Output: TTS systems benchmarked on prosody, intelligibility, and naturalness with language-specific evaluation criteria

Challenge 9: Long-Form TTS

- Goal: Generate extended continuous speech — audiobook narration, news reading, educational content — that maintains consistent quality, prosody, speaker identity, and natural pacing over minutes of output
- Output: TTS systems evaluated on prosodic coherence, breath and pause naturalness, and quality degradation over long-form Indian language passages

Speech-to-Speech Translation (S2S)

Challenge 10: Techniques to Construct S2S Datasets

- Goal: Develop scalable techniques for building parallel speech datasets for Indian language pairs — through synthetic generation, alignment of existing corpora, mining multilingual broadcast content, or bootstrapping from text translation pairs
- Output: Dataset creation pipelines and resulting datasets with documented quality, diversity, and coverage across Indian language pairs

Challenge 11: Building S2S Models — Cascaded vs End-to-End

- Goal: Build and compare cascaded (ASR → MT → TTS) and end-to-end spoken translation systems on Indian language pairs, evaluating trade-offs in accuracy, latency, naturalness, and robustness
- Output: Trained S2S models with comparative benchmarks across both paradigms on Indian language pairs

Challenge 12: Evaluation Metrics for S2S

- Goal: Propose evaluation frameworks that holistically measure translation fidelity, fluency, prosodic quality, and latency for Indian language S2S — beyond simply running ASR on output and computing BLEU
- Output: New S2S evaluation metrics, toolkits, and validation studies for Indian language pairs

Spoken Question Answering

Challenge 13: Designing Evaluation Metrics for Spoken QA

- Goal: Develop evaluation metrics that assess spoken QA systems on answer correctness, robustness to ASR errors and disfluencies, handling of code-switched questions, and quality differences between MCQ and descriptive answers
- Output: Evaluation protocols and metric implementations tailored to Indian language spoken QA

Challenge 14: Spoken QA Model Building — Cascaded vs End-to-End

- Goal: Build and compare cascaded (ASR → text QA) and end-to-end spoken QA systems on Indian languages, testing whether end-to-end models preserve information lost in the ASR step
- Output: Trained spoken QA models with comparative benchmarks on accuracy, latency, and robustness to spoken language phenomena

Challenge 15: Dataset Creation Techniques for Spoken QA

- Goal: Develop scalable methods for constructing spoken QA datasets for Indian languages — synthetic question generation, crowdsourced recording, TTS-augmented data, or extraction from educational and media content
- Output: Dataset creation pipelines and resulting spoken QA datasets across Indian languages and domains

Challenge 16: MCQ vs Descriptive Spoken QA

- Goal: Separately evaluate systems on MCQ-based spoken QA (selecting from options) and descriptive spoken QA (generating free-form answers) to understand where systems genuinely comprehend versus rely on surface-level matching
- Output: Systems benchmarked on both formats with analysis of performance gaps between MCQ and descriptive tasks

Speech Language Models (Speech LLMs)

Challenge 17: Building Speech LLMs for Indian Languages

- Goal: Build Speech LLMs that operate directly on speech representations for Indian languages — handling spoken instruction following, speech-to-speech dialogue, spoken summarization, and multilingual speech understanding
- Output: Speech LLM models evaluated across Indian languages on reasoning quality, generation coherence, and robustness to accent and dialect variation

Challenge 18: Speaker Characteristics from Speech LLMs

- Goal: Explore whether Speech LLMs can extract and utilize speaker-related characteristics — age, gender, emotion, accent, speaking style — for Indian language speakers, spanning data requirements, methods, and evaluation
- Output: Techniques, datasets, and evaluation frameworks for speaker characteristic extraction from Speech LLMs, with analysis of accuracy and fairness

Data

Challenge 19: Rich Transcription Data Creation

- Goal: Create datasets with annotations that go beyond plain orthographic text — speaker labels, language tags, prosodic markers, disfluency annotations, domain labels, and emotional cues for Indian speech
- Output: Rich transcription datasets and annotation guidelines for Indian ASR, TTS, and Spoken QA

Challenge 20: Data Creation with Minimal Human Effort

- Goal: Develop end-to-end pipelines for creating high-quality labeled speech data with minimal human annotation — using pseudo-labeling, active learning, forced alignment, synthetic generation, LLM-assisted annotation, or web-crawled audio with automatic filtering
- Output: Documented, reproducible data creation pipelines with quality benchmarks demonstrating viability for training competitive Indian language systems

Language Challenges

Track 1 : Robust MT for Indian Languages

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Lab : PLURAL (Param)

Subtasks:

- 1) Challenge Set Evaluation
- 2) Dialect Robustness
- 3) Error Analysis Track System

Idea :

1. Develop linguistically grounded diagnostic challenge sets for Indian languages that go beyond generic test sets and explicitly evaluate machine translation

systems on specific linguistic phenomena such as morphology, syntax, code-switching, idiomatic expressions, and domain-specific usage. These challenge sets will consist of carefully curated examples (including minimal pairs) designed to isolate and test individual capabilities of MT systems, enabling fine-grained evaluation of model behavior rather than relying on aggregate scores like BLEU.

2. Create and evaluate MT systems on dialectally diverse inputs across Indian languages, capturing real-world linguistic variation that is typically absent in standard training and evaluation datasets.

This includes:

- Regional dialects (e.g., Telangana Telugu, Kongu Tamil)
- Informal and spoken variants
- Non-standard grammar and lexical variation

The task aims to assess:

- Whether MT systems can generalize beyond standardized forms
 - Whether they fail, normalize incorrectly, or mistranslate dialectal inputs
3. Introduce a structured error analysis framework for Indian MT, requiring participating systems to go beyond output generation and provide fine-grained error categorization and analysis.

Participants will analyze system outputs using a standardized error taxonomy, including:

- Morphological errors (agreement, inflection)
- Syntactic errors (word order, clause structure)
- Lexical errors (word choice, ambiguity)
- Named entity and transliteration errors
- Dialect misinterpretation

The goal is to:

- Encourage interpretability and transparency in MT systems
 - Enable systematic comparison of strengths and weaknesses
 - Build a shared understanding of common failure modes in Indian MT
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What can we target for EWSLT 2026 ?

Track 1: Spoken Question Answering (SQA)

Designing Evaluation Metrics

- Develop automatic evaluation metrics for SQA that demonstrate high correlation with human judgments of answer quality
- Incorporate statistical rigor in reporting — appropriate significance tests, score distributions rather than single-point estimates, and confidence intervals to ensure trustworthiness of results and avoid chance findings

Creating Baseline Models

- End-to-End: Build models that directly take spoken questions as input and produce answers without intermediate text representations
 - Cascaded pipelines:
 - ASR → LLM: Transcribe spoken question, then pass to a language model for answer generation
 - ASR → MT → LLM: Transcribe, translate to a pivot language if needed, then generate answers — relevant for low-resource languages where LLM performance is stronger in another language
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Track 2: Evaluating Speech-to-Speech Translation (S2S) / Text-to-Speech (TTS)

Designing Evaluation Metrics for S2S / TTS

- Develop automatic metrics that achieve high correlation with human judgments of translation and synthesis quality
- Reference-free metrics: Evaluation measures that require only the source speech and system predictions — no human reference translations or reference audio needed
- Cross-lingual metrics: Measures that can accept inputs in different languages, enabling evaluation across diverse Indian language pairs without language-specific metric tuning
- Statistical rigor in reporting: Appropriate significance tests, score distributions rather than single-point estimates, and confidence intervals to ensure trustworthiness and avoid chance findings

Creating Baseline Models for S2S

- End-to-End: Build models that directly translate speech in one Indian language to speech in another without intermediate text
- Cascaded: ASR → MT → TTS pipeline as a baseline for comparison against end-to-end approaches

TTS Evaluation (if TTS track is included)

- Naturalness: How human-like does the synthesized speech sound?

- Prosody: Does the system capture correct intonation, rhythm, stress, and phrasing patterns specific to the target Indian language?
- Intelligibility: Can listeners accurately understand the synthesized speech content?